Computer Vision Practical Report:
GRADIENT TECHNIQUE

1. AIM

To compute the Morphological Gradient (the difference between dilation and erosion of an image) to highlight structural boundaries and object outlines using cv2.morphologyEx() in OpenCV, display the results, and save the output image.

2. PROCEDURE

- Load the input image using cv2.imread() and convert it to grayscale using cv2.cvtColor().
- Define a spatial structuring element (kernel) using cv2.getStructuringElement() (e.g., rectangular or elliptical).
- Apply the morphological gradient operation using cv2.morphologyEx() with cv2.MORPH_GRADIENT.
- Display the original and morphological gradient images side-by-side using Matplotlib.
- Save the output image to disk using cv2.imwrite().

3. PROGRAM CODE

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Load image
img_bgr = cv2.imread('E33-IMG.jpg')
img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)

# 2. Define Structuring Element (5x5 kernel)
kernel = np.ones((5, 5), np.uint8)

# 3. Perform Morphological Gradient (Dilation - Erosion)
gradient = cv2.morphologyEx(img_rgb, cv2.MORPH_GRADIENT, kernel)
```

```
# 4. Display side-by-side on a SINGLE PAGE
fig, axes = plt.subplots(1, 2, figsize=(8, 4))

axes[0].imshow(img_rgb)
axes[0].set_title('Original Image')
axes[0].axis('off')

axes[1].imshow(gradient)
axes[1].set_title('Morphological Gradient')
axes[1].axis('off')

plt.tight_layout()
plt.show()
```

4. OUTPUT

5. RESULT

The Python script executed successfully. The input image was loaded.